

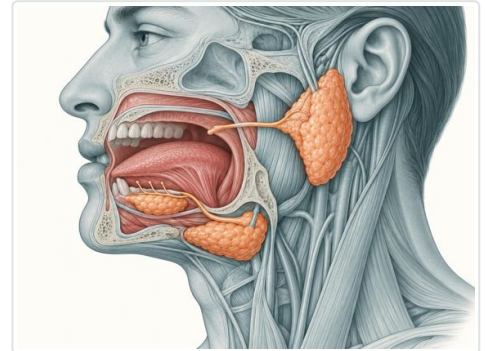
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which cranial nerve drives the parotid gland, and which drives the submandibular and sublingual glands?
2. Which phases of swallowing are voluntary, and which are automatic?
3. Dysphagia for solids only versus solids and liquids — which points to mechanical obstruction, and which to a motility disorder?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Major salivary glands

Gland	Secretion	Key point
Parotid	Mostly serous , amylase-rich	Contribution rises strongly with stimulation
Submandibular	Mixed serous + mucous	Major source of unstimulated saliva
Sublingual	Mostly mucous	Important for lubrication of the oral cavity
Minor glands	—	Continuously support mucosal lubrication



Major salivary glands — location predicts secretion.

2 What saliva contains and does

Function	Key players	What it does
Lubricate	Mucins	Coat food and mucosa → easier speech, chewing and swallowing
Digest	α-amylase , lingual lipase	Amylase begins starch digestion ; lingual lipase contributes to fat digestion
Taste	Water (solvent)	Dissolves molecules so they can reach taste receptors
Buffer	Bicarbonate	Neutralises acids; flow-dependent protection after meals
Protect	IgA, lysozyme, lactoferrin, peroxidases, antimicrobial peptides	Mechanical cleansing + antimicrobial defence of teeth and mucosa
Teeth	Ca ²⁺ , phosphate	Remineralisation ; less saliva → more dental caries

- Composition: **~99% water**; Na⁺, Cl⁻, K⁺, HCO₃⁻; α-amylase, mucins, IgA, lysozyme, lactoferrin; urea, uric acid, growth factors.

ACINUS → DUCT. Acini make a **primary isotonic** fluid; the ducts modify it into **mostly hypotonic** saliva. **Higher flow = less ductal modification.**

3 Control of salivation

	Parasympathetic	Sympathetic
Pathway	CN VII → submandibular + sublingual CN IX → parotid	Thoracic sympathetic outflow → superior cervical ganglion → glands
Saliva	Copious, watery	Smaller-volume, protein-rich
Also	Marked vasodilation + ↑ glandular blood flow	Myoepithelial contraction helps expulsion; strong sympathetic states can feel “dry”

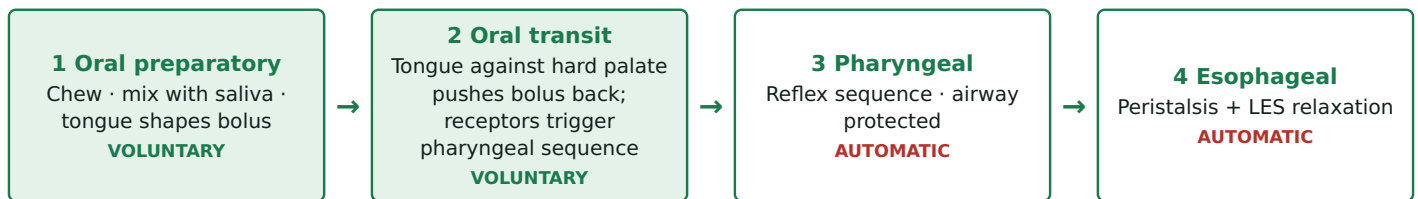
↑ Salivation	↓ Salivation
Sour taste	Sleep — low basal output
Chewing — tactile oral input	Dehydration — less fluid available
Smell + sight — conditioned responses	Fear / anxiety — dry-mouth sensation
Nausea — marked salivation	Antimuscarinic drugs — block cholinergic drive

CEPHALIC PHASE. Saliva can rise **before food enters the mouth**. Brainstem salivatory nuclei send the autonomic output to the glands. Both divisions stimulate salivation — quantity and composition differ.

4 Swallowing: three tasks, four phases

- Every swallow must **move the bolus** (mouth → pharynx → esophagus), **protect the airway** (pause breathing, close the laryngeal inlet, prevent aspiration) and **open the right sphincters** (UES opens, LES relaxes). **Timing is everything.**

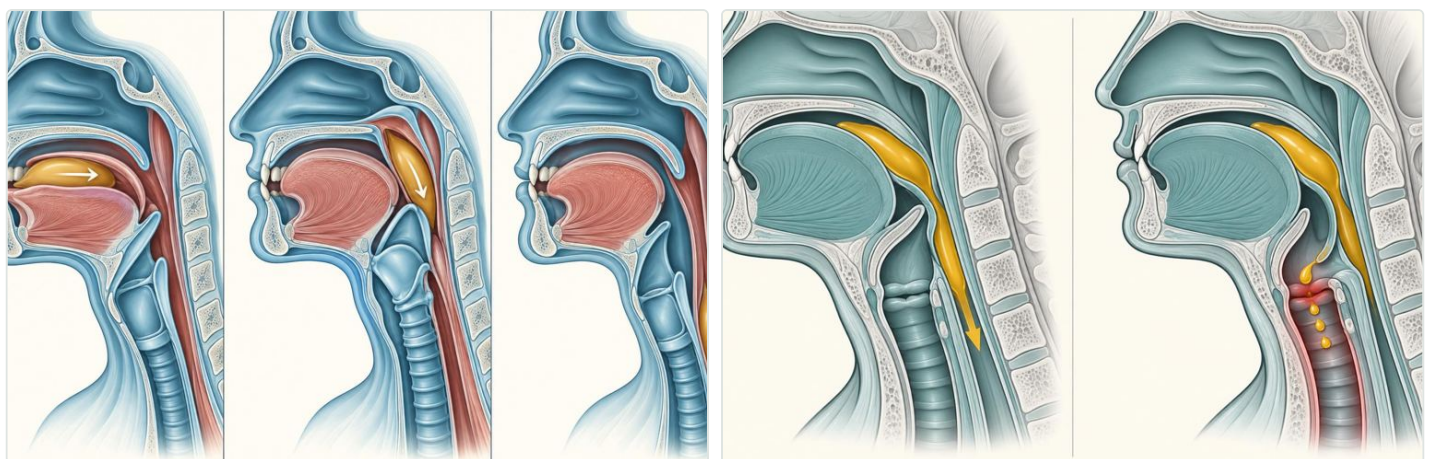
VISUAL ANCHOR · THE FOUR PHASES — voluntary → automatic at the oropharynx



5 Pharyngeal phase and airway protection

Pharyngeal phase — rapid, programmed	Airway protection is layered
Soft palate elevates → nasopharynx closes Larynx elevates and moves anteriorly Vocal folds adduct; epiglottis tilts UES relaxes Pharyngeal constrictors propel the bolus - Breathing pauses briefly: deglutition apnea	True and false vocal folds close Arytenoids approximate Larynx elevates and moves forward Epiglottis redirects the bolus Cough reflex responds if material enters the airway

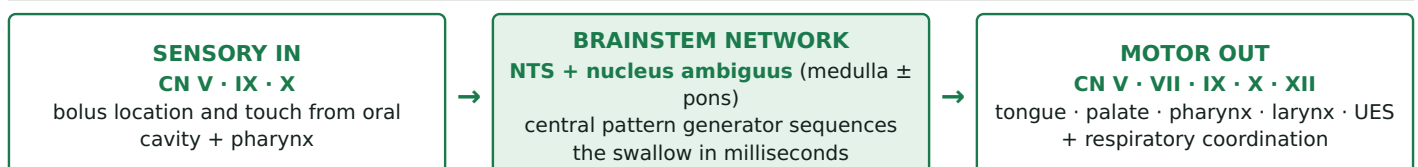
NEVER MISS. The **epiglottis alone does NOT “seal” the airway** — protection comes from several layers working together.



Bolus enters pharynx · airway closes · bolus enters esophagus

Safe swallow (left) vs aspiration (right)

6 Neural control of swallowing



7 Esophageal phase

Feature	Detail
Primary peristalsis	Continuation of the swallowing wave
Secondary peristalsis	Triggered by esophageal distension if material remains
Gravity	Helps — but is not required
LES	Relaxes before the bolus arrives (deglutitive inhibition via vagal + enteric inhibitory pathways, NO / VIP); tone returns after the bolus passes

8 Clinical: localise the problem from the history

	Oropharyngeal	Esophageal
Patient says	"I cannot start the swallow."	"Food sticks after I swallow."
Clues	Coughing/choking immediately , nasal regurgitation, wet voice; neurologic disease often present	Bolus sensation behind the sternum ; usually no immediate choking at initiation
Discriminator	Solids only → think mechanical obstruction · Solids + liquids → think motility disorder	
Problem zone	Oral/pharyngeal control + airway protection	Esophageal lumen, peristalsis or LES

Lecture case	Physiology
62-year-old, dry mouth on an antimuscarinic; crackers hard to swallow, taste ↓, caries ↑	↓ muscarinic stimulation → ↓ watery flow → ↓ lubrication + solvent for taste → ↓ buffering / cleansing / remineralisation
35-year-old, pain + swelling under the jaw, worse when eating	Meal stimulus ↑ secretion against an obstructed duct (stone) → back-pressure → painful gland distension
73-year-old after stroke, coughs immediately after water, wet gurgly voice	Oropharyngeal/pharyngeal dysphagia : failed timing or strength of airway protection + pharyngeal propulsion → aspiration risk (red flag)
Achalasia	Loss of inhibitory myenteric neurons (NO + VIP) → LES fails to relax + absent/abnormal peristalsis → solids and liquids both difficult

P Must-remember pearls

- Acini make a **primary isotonic** fluid; ducts modify it to **mostly hypotonic** saliva; **higher flow = less ductal modification**.
- **Parasympathetic** (CN VII → submandibular + sublingual, CN IX → parotid) = copious watery saliva; **sympathetic** (via superior cervical ganglion) = smaller, protein-rich, can feel dry.
- **Oral phases are voluntary; pharyngeal + esophageal are automatic** — the handover happens as the bolus reaches the oropharynx.
- Airway protection is **layered**; the **epiglottis alone does not seal the airway**. Breathing pauses briefly (deglutition apnea).
- The LES relaxes **ahead of the bolus via NO/VIP**; lose this and you get achalasia. **Solids only = mechanical; solids + liquids = motility**.

Q Test yourself

1. A 58-year-old starts a drug that blocks muscarinic receptors. Soon he complains of a dry mouth and finds dry crackers hard to swallow. Which change best explains the swallowing difficulty?

- A. Failure of the upper esophageal sphincter to relax
- B. Loss of deglutition apnea, so the airway is unprotected
- C. Less watery saliva, because cholinergic drive to the glands is blocked and lubrication falls
- D. Loss of the sympathetic supply via the superior cervical ganglion, so no saliva is made
- E. Loss of α -amylase, so starch can no longer be swallowed

2. While eating, a man deliberately chews and pushes the bolus back with his tongue. Which event marks the point after which the swallow proceeds without voluntary control?

- A. The bolus reaching the oropharynx, which triggers the pharyngeal reflex sequence
- B. The bolus being shaped by the tongue during chewing
- C. The LES relaxing ahead of the bolus
- D. The bolus arriving in the stomach
- E. The bolus leaving the lips

3. During a swallow, which combination correctly describes the events of the pharyngeal phase?

- A. Soft palate lowers, larynx elevates and moves anteriorly, vocal folds adduct, UES relaxes
- B. Soft palate elevates, larynx elevates and moves anteriorly, vocal folds adduct, UES contracts
- C. Soft palate elevates, larynx elevates and moves anteriorly, vocal folds abduct, UES relaxes
- D. Soft palate elevates, larynx elevates and moves anteriorly, vocal folds adduct, UES relaxes
- E. Soft palate elevates, larynx descends, vocal folds adduct, UES relaxes

4. A patient describes food “sticking” behind the sternum after every swallow, for both solids and liquids, with no choking when he starts to swallow. Which is the most likely physiological problem?

- A. Reduced salivary flow from an antimuscarinic drug
- B. Failure of the soft palate to close the nasopharynx
- C. Failed airway protection at the pharyngeal phase
- D. A mechanical obstruction of the esophageal lumen
- E. An esophageal motility disorder, such as achalasia

5. A 35-year-old man has recurrent pain and swelling under the jaw that is worse when he eats. Which mechanism best explains why eating makes it worse?

- A. Sympathetic overactivity causes the myoepithelial cells to relax and the gland to swell
- B. Eating drives secretion against an obstructed duct, causing back-pressure and painful distension
- C. The LES fails to relax, so food is retained behind the sternum
- D. Saliva is aspirated through the open laryngeal inlet
- E. Eating reduces salivary flow, so the gland becomes dry and inflamed

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — C — Antimuscarinics block cholinergic drive → ↓ watery flow → ↓ lubrication (and solvent for taste, buffering, remineralisation).

2 — A — Voluntary control ends as the bolus reaches the oropharynx; the pharyngeal and esophageal phases are automatic.

3 — D — Palate up (nasopharynx closed), larynx up and forward, vocal folds adducted, UES relaxed; constrictors propel the bolus and breathing pauses.

4 — E — Solids + liquids → motility disorder. Achalasia = loss of inhibitory myenteric neurons (NO/VIP) → LES fails to relax.

5 — B — Lecture case: meal/taste stimulus ↑ secretion → stone in duct → back-pressure → painful distension.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Physiology GIT Block · MEDucation by Mattin Majid · GIT LG 1 & 2 slides, 2026-2027

Study alongside the lecture, not instead of it.